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PROBLEM STATEMENT - 3

Traditional file management relies on rigid, manual hierarchies of folders and sub-folders that quickly become disorganised and inefficient. The Semantic Entropy File System (SEFS) challenge tasks you with developing a self-organising file manager that replaces static directories with a dynamic semantic layer integrated directly into the operating system. Instead of requiring users to categorise files manually, the system must autonomously analyse the content of documents and project them into a live file structure (for example: Graph, FlowChart, Spatial Map, etc.). In this environment, files with high conceptual similarity automatically converge into meaningful folders, while unrelated documents remain separated.

All semantic processing must operate on files located within a single monitored root folder. Within this root directory, the system must dynamically organise files into multiple semantic folders based on their content. Each folder should contain only the files that conceptually belong together.

Unlike conventional file managers, this semantic structure must function at the OS level rather than as a standalone visualisation tool. The system should monitor file operations such as creation, modification, renaming, and deletion, and continuously synchronise semantic organisation with the operating system’s file view. When semantic relationships change, the OS-level folder structure must update accordingly.

The core difficulty lies in maintaining a “living” semantic layout: as new data is introduced or existing content changes, the system must re-calculate file positioning, update the OS-integrated folder structure, and apply live changes in real time without breaking file integrity or disrupting normal system operations.

Expected Outcomes

1. Auto-Detection & Processing: A system that monitors a single root directory and automatically processes any new or modified PDF or text file added to the operating system.

2. Semantic Folder Organisation: Content-based logic that dynamically creates and maintains multiple folders inside the root directory, ensuring each folder contains only semantically related files.

3. OS-Level Synchronisation: Bidirectional consistency where semantic reorganisation updates the operating system's folder structure, and any manual OS-level file action triggers semantic re-analysis.

4. Interactive Interface with Live Changes: A visual 2D interface where files are represented as nodes, users can hover for metadata and open files, and where the layout reflects live changes whenever semantic recalculation occurs.